CONTINUOUS RANDOM VARIABLE

1. A continuous random variable X has the following density function, $f(x) = kx(2-x)$ for $0 < x < 2$, find the value of (i) k (ii) Mean of X (iii) Variance of X.
2. A continuous random variable X has the following density function, $f(x) = cx^2$ for $|x| < 1$. find the value of (i) c (ii) Mean of X (iii) Variance of X (iv) $P\left(X \geq \frac{1}{2}\right)$.
3. Find the expected value of a continuous random variable that follows a pdf given by $f(x) = \begin{cases} 3x & 0 \leq x \leq 1 \\ x+1 & 1 < x \leq 2 \\ 0 & otherwise \end{cases}$

UNIFORM (OR RECTANGULAR) DISTRIBUTION

1. Let X be a random variable following a uniform distribution $X \sim U(a,b)$. Show that, the mean or expected value of X is $E(x) = \mu = \frac{a+b}{2}$.
2. Let X be a random variable following a uniform distribution $X \sim U(a,b)$ Then, show that, the variance of X is $v(x) = \sigma^2 = \frac{(b-a)^2}{12}$.
3. A bus arrives at a bus stop every 20 minutes. If you arrive at the bus stop at random time, what is the probability that you will have to wait more than 15 minutes for the bus?
4. A machine produces parts that have a thickness uniformly distributed between 1.5 and 2.5 *mm*. what is the probability that a part has a thickness between 1.8 *mm* and 2.2 *mm*?
5. Bus is uniformly late between 2 and 10 minutes. How long can you expect to wait? With what standard deviation? If there is a delay of more than 7 minutes, you will be late for work. What is the probability of you being late?
6. If X is uniformly distributed with mean 1 and variance $\frac{4}{3}$, find $P(X < 0)$.
7. Subway trains on a certain line run every half hour between mid-night and six in the morning. What is the probability that man entering the station at a random time during this period will have to wait at least 20 minutes?
8. Ace Heating and Air Conditioning Service finds that the amount of time a repairman needs to fix a furnace is uniformly distributed between 1.5 and four hours. Let the time needed to fix a furnace. Then.

a) Find the probability that a randomly selected furnace repair requires more than two hours.
b) Find the probability that a randomly selected furnace repair requires less than three hours.
c) Find the 30th percentile of furnace repair times.
d) The longest 25% of furnace repair times take at least how long? (In other words: find the minimum time for the longest 25% of repair times.) What percentile does this represent?
e) Find the mean and standard deviation.

EXPONENTIAL DISTRIBUTION

1. Let X be a random variable following an exponential distribution $X \sim exp(\theta)$. Show that, the mean or expected value of X is $E(x) = \mu$.

2. Let X be a random variable following an exponential distribution $X \sim exp(\theta)$ Then, show that, the variance of X is $v(x) = \mu^2$.

3. The lifetime of a light bulb follows an exponential distribution with a mean of 1000 hrs. what is the probability that a light bulb will last more than 1500 hrs.?

4. The time between failures of a machine follows an exponential distribution with a mean of 50 hrs. what is the probability that, the machine will fail within the next 20 hrs.?

5. The lifetime of a battery follows exponential distribution with mean of 200 hrs. what is the probability that, the battery will last for more than 300hrs.?

6. On the average, a certain computer part lasts ten years. The length of time the computer part lasts is exponentially distributed. (a) What is the probability that a computer part lasts more than 7 years? (b) What is the probability that a computer part lasts between 9 and 11 years?

7. The number of days ahead travelers purchase their airline tickets can be modeled by an exponential distribution with the average amount of time equal to 15 days. Find the probability that a traveler will purchase a ticket fewer than ten days in advance.?

8. Let X = amount of time (in minutes) a postal clerk spends with his or her customer. The time is known to have an exponential distribution with the average amount of time equal to four minutes. Find the probability that a clerk spends four to five minutes with a randomly selected customer?

9. The mileage which car owners get with a certain kind of radial tyre is a random variable having an exponential distribution with mean 40,000 km. Find the probabilities that one of these tires will last (i) at least 20,000 km and (ii) at most 30,000 km.?

10. The length of time a person speaks over phone follows exponential distribution with mean 6. What is the probability that the person will talk for (1) more than 8 minutes (2) between 4 and 8 minutes?

NORMAL DISTRIBUTION

1. Let X be a random variable following a normal distribution $X \sim N(\mu, \sigma^2)$. Show that, the mean or expected value of X is $E(x) = \mu$.

2. Let X be a random variable following a normal distribution $X \sim N(\mu, \sigma^2)$. Show that, the variance of X is $v(x) = \sigma^2$.

3. In a sample of 1000 candidates, the mean of certain test is 14 and standard deviation is 2.5. Assuming Normal distribution, find the number of candidates getting less than eight marks. $[Given\ that\ z = 2.4, A = 0.4918]$.

4. In a normally distributed group of 450 students with mean 42 and standard deviation 8, find the number of students scoring less than 48 marks. $[Given\ that\ z = 0.75, A = 0.2734]$.

5. In a certain examination test 10000 students appeared in a paper of mathematics. Average marks obtained were 50% with Standard Deviation 5%. Marks are normally distributed. Find the number of students expected to get more than 60% marks. $[Given\ z = 2, A = 0.4772]$.

6. 5000 candidates appeared in a certain paper carrying a maximum of 100 marks. It was found that marks were normally distributed with mean 39.5 and standard deviation 12.5. Determine the number of candidates who secured a first class for which a minimum of 60 marks is necessary. $[Given\ that\ z = 1.64, A = 0.4495]$.

7. A random sample of 200 screws is drawn from a population which represents the size of screws. If a sample is distributed normally with mean 3.15 cm and standard deviation 0.025 cm, find expected number of screws whose size falls between 3.12 cm and 3.2 cm. $[Given\ that\ [z = 1.2, A = 0.3849]\ and\ [z = 2, A = 0.4772]$.

8. The mean weight of 500 students is 63kg and standard deviation is 8kg. Assuming that the weights are normally distributed, find how many students weigh 52kg? The weights are recorded to the nearest kg.
[$Given\ that, z = 1.44, A = 0.4251\ and\ for\ z = 1.31, A = 0.4049$].

9. Assuming that, the diameter of 1000 brass plugs taken consecutively from machine form a normal distribution with mean 0.7515 cm and standard deviation 0.002 cm. How many of the plugs are likely to be appeared if the acceptable diameter is 0.752 ± 0.004.
[$Given\ that, z = 2.25, A = 0.4878\ and\ z = 1.75, A = 0.4599$].

10. Suppose heights of student's follows normal distribution with mean 190cm and variance 80 cm. In a school of 1000 students how many would you expect to be above 200 a tall?
[$Given\ that\ z = 1.118, A = 0.3686$].

11. The lifetime of an article has a normal distribution with mean 400 hours and standard deviation 50 hours. Assuming normal distribution, find the expected number of articles out of 2000 whose lifetime lies between 335 hours to 465 hours? [$Given: z = 1.3, A = 0.4032$]

12. The lifetime of a certain component has a normal distribution with mean 400 hours and standard deviation 50 hours. Assuming normal distribution, determine approximately the expected number of components in a sample of 1,000 components whose lifetime lies between 340 hours to 465 hours? [$Given: z1 = 1.2, A1 = 0.3849;\ z2 = 1.3, A2 = 0.4032$]

13. In a sample of 1,000 cases, the mean of a certain examination is 14 and standard deviation is 2.5. Assuming the distribution to be normal, find the number of students scoring between 12 and 15. [$Given: z1 = 0.4, A1 = 0.1554;\ z2 = 0.8, A2 = 0.2881$]

14. In a distribution, exactly normal, 7% of the items are under 35 and 89% are under 63. Find the mean and standard deviation of the distribution.
[Given that $Area\ A_1 = P(0 < z < 1.48) = 0.43$ and $Area\ A_2 = P(0 < z < 1.23) = 0.39$]

15. In a distribution, exactly normal, 10% of the items are under 40 and 5% are under 80. Find the mean and standard deviation of the distribution.
[Given that $Area\ A_1 = P(0 < z < 1.29) = 0.4$ and $Area\ A_2 = P(0 < z < 1.65) = 0.45$].

16. If the random variable X is normally distributed with mean 80 and standard deviation 5, then find (a) $P(X > 95)$ (b) $P(X < 72)$ (c) $P(60.5 < X < 90)$
(d) $P(85 < X < 97)$ (e) $P(64 < X < 76)$

[Given that $A = P(z = 3) = 0.4987, A = P(z = 1.6) = 0.4452,\ A = P(z = 3.9) = 0.5,$
$A = P(z = 2) = 0.4772,\ A = P(z = 3.4) = 0.4997, A = P(z = 1) = 0.3413$
$A = P(z = 3.2) = 0.4993, A = P(z = 0.8) = 0.2881$].

CHI-SQUARE DISTRIBUTION

1. In experiment on pea breading, the following frequencies of seeds were obtained.

Round and Green	Wrinkled and Green	Round and Yellow	Wrinkled and Yellow	Total
222	120	32	150	524

Theory predicts that the frequencies should be in proportion 8:2:2:1. find χ_3^2.

2. Among 64 offspring of a certain cross between guinea pigs 34 were red, 10were black and 20 were white. According to genetic model, these numbers should be in the ratio 9:3:4. Find χ^2.

3. A sample analysis of examination results of 500 students was made. The observed frequencies are 220, 170, 90, 20 and the numbers are in the ratio 4:3:2:1 Find χ^2.